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Project Proposal: Resume Optimizer and Interview Prep Assistant

1. Problem Statement and Motivation

a. What real-world business problem are you addressing?

The job market is becoming more competitive and AI-reliant every day, which makes it difficult for college students and those in the early stages of their career journeys. Applicant tracking systems, keyword filters, and other AI-reliant hiring processes evaluate applicants first, making it harder for student applications to reach human recruiters. Due to this shift in hiring practices, many students do not hear back from their applications, and they may feel lost and wonder how they can improve.

The root of the problem comes from students being unsure of how to phrase and present their experience to match the job requirements in a way that AI can detect and understand. Applicants may have all the required classwork, projects, and experience, but if they do not optimize their resume for the preliminary AI screening, they will be taken from the candidate pool early on. Due to this, many applicants may feel discouraged and unprepared.

b. Who are the target users?

The target users are college students, recent graduates, internship and entry-level job applicants, and those switching career fields.

c. Why is an agentic AI solution appropriate here (vs. a human or an external party defining the workflow)?

My agentic AI solution for this issue is HireFit AI. HireFit AI will provide personalized support to students on their career search journey by helping them understand how well their experience aligns with a specific job posting's requirements, optimizing the phrasing and format

of their resume to the posting, and creating a short list of potential interview questions for the user to practice. HireFit AI will provide convenient, remote, and flexible personalized help that in-person career centers are not always able to provide. HireFit AI is also a good option for those who feel intimidated or shy about getting resume feedback from a human reviewer.

2. Use cases & scenarios

a. Use case 1: Resume to Job Match Score

A student wants to see how many of the job qualifications they meet. They upload their resume and the job posting.

Agent tasks:

- Scan the resume and job description
- Find overlapping skills and tools
- Create a compatibility score
- List any missing qualifications

b. Use case 2: Resume Improvement

A student wants to tailor their resume to a specific job.

Agent tasks:

- Rewrite bullet points to include job keywords
- Emphasize the impact of experiences and improve clarity

c. Use case 3: Interview preparation

A student wants to practice for an interview.

Agent tasks:

- Create specific interview questions tailored to a job posting
- Give an example of an ideal answer

- Give feedback from a recruiter's viewpoint

3. Initial system design

Agents and roles:

- Resume Analyzer: extracts skills and experience
- Job Market Researcher: collects job requirements
- Job-Resume Compatibility Agent: compares resume to job requirements
- Resume Writer: tailors resume content
- Interview Coach: creates questions and feedback
- Recruiter Critic: reviews the accuracy of responses

Expected tools and APIs:

- Document parser
- Similarity scoring
- LLM API
- Job posting scraper
- Front-end interface

4. Feasibility & risks

a. Potential challenges (e.g., hallucination, long context, tool failures).

- Hallucinated skills/experiences
- Long documents exceeding the character limit
- Tool/API failures
- Privacy concerns with resume uploads

b. Any constraints (e.g., rate limits, privacy constraints, computation).

- API rate limit

- Computation limits
 - User data protection
- c. Timeline for project execution (cannot extend beyond the class duration).

Week	Task
1	Proposal and system design
2	Front-end prototype
3	Multi-agent workflow
4	Evaluation and pilot
5	Final report and presentation

5. Conclusion:

HireFit AI is an agentic AI tool that makes the job search process less intimidating for students. Through the use of multiple agents that work together and critique each other, HireFit provides career guidance to users. This tool will show how agentic AI can solve a real business problem.